

10. The tables show the molecular formulae of some alkanes and alkenes.

Alkanes	Alkenes
CH_4	C_2H_4
C_2H_6	C_3H_6
C_3H_8	
C_4H_{10}	

- (a) The general formula for the alkene family is C_nH_{2n} . Give the general formula for the alkane family. [1]

.....

- (b) When alkanes and alkenes completely burn in air they form the same two products. Give the chemical formulae for both products. [1]

..... and

- (c) Draw the structural formula for propene. [1]

- (d) Bromine water is used to distinguish alkenes from alkanes. Describe the colour **change** seen when bromine water is added to an alkene. [1]

.....

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



2
1

Group

1 H Hydrogen 1										4 He Helium 2									
7 Li Lithium 3		9 Be Beryllium 4		23 Na Sodium 11		24 Mg Magnesium 12		39 K Potassium 19		40 Ca Calcium 20		86 Rb Rubidium 37		133 Cs Caesium 55		223 Fr Francium 87			
11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17			
27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18		70 Ga Gallium 31		73 Ge Germanium 32		80 Br Bromine 35			
51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		84 Kr Krypton 36			
81 Tl Thallium 81		82 Pb Lead 82		83 Bi Bismuth 83		84 Po Polonium 84		85 At Astatine 85		86 Rn Radon 86		115 In Indium 49		119 Sn Tin 50		127 I Iodine 53			
137 La Lanthanum 57		139 Y Yttrium 39		141 Nb Niobium 41		143 Ta Tantalum 73		145 Hf Hafnium 72		147 Zr Zirconium 40		150 Mo Molybdenum 42		153 Tc Technetium 43		156 Ru Ruthenium 44			
181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Ti Thallium 81			
226 Ra Radium 88		227 Ac Actinium 89		228 Th Thorium 90		231 Pa Protactinium 91		232 U Uranium 92		235 Pu Plutonium 94		238 Am Americium 95		241 Cm Curium 96		244 Bk Berkelium 97			
287 Nh Nihonium 113		288 Ds Darmstadtium 114		289 Ts Tennessine 115		290 Og Oganesson 116		291 Nh Nihonium 113		292 Ds Darmstadtium 114		293 Ts Tennessine 115		294 Og Oganesson 116		295 Nh Nihonium 113			
101 Md Mendelevium 101		102 No Nobelium 102		103 Lr Lawrencium 103		104 Rf Rutherfordium 104		105 Db Dubnium 105		106 Sg Seaborgium 106		107 Bh Bohrium 107		108 Hs Hassium 108		109 Mt Meitnerium 109			
111 Rg Roentgenium 111		112 Uub Ununbium 112		113 Uut Ununtrium 113		114 Uuq Ununquadium 114		115 Uup Ununpentium 115		116 Uuh Ununhexium 116		117 Uus Ununseptium 117		118 Uuo Ununoctium 118		119 Uut Ununtrium 113			
121 Ubu Unbibium 121		122 Ubb Unbibium 122		123 Ubc Unbium 123		124 Ubd Unbibium 124		125 Ube Unbeium 125		126 Ubf Unbibium 126		127 Ubg Unbium 127		128 Ubh Unbeium 128		129 Ubi Unbeium 129			
131 Ubu Unbibium 131		132 Ubb Unbibium 132		133 Ubc Unbium 133		134 Ubd Unbibium 134		135 Ube Unbeium 135		136 Ubf Unbibium 136		137 Ubg Unbium 137		138 Ubh Unbeium 138		139 Ubi Unbeium 139			
141 Ubu Unbibium 141		142 Ubb Unbibium 142		143 Ubc Unbium 143		144 Ubd Unbibium 144		145 Ube Unbeium 145		146 Ubf Unbibium 146		147 Ubg Unbium 147		148 Ubh Unbeium 148		149 Ubi Unbeium 149			
151 Ubu Unbibium 151		152 Ubb Unbibium 152		153 Ubc Unbium 153		154 Ubd Unbibium 154		155 Ube Unbeium 155		156 Ubf Unbibium 156		157 Ubg Unbium 157		158 Ubh Unbeium 158		159 Ubi Unbeium 159			
161 Ubu Unbibium 161		162 Ubb Unbibium 162		163 Ubc Unbium 163		164 Ubd Unbibium 164		165 Ube Unbeium 165		166 Ubf Unbibium 166		167 Ubg Unbium 167		168 Ubh Unbeium 168		169 Ubi Unbeium 169			
171 Ubu Unbibium 171		172 Ubb Unbibium 172		173 Ubc Unbium 173		174 Ubd Unbibium 174		175 Ube Unbeium 175		176 Ubf Unbibium 176		177 Ubg Unbium 177		178 Ubh Unbeium 178		179 Ubi Unbeium 179			
181 Ubu Unbibium 181		182 Ubb Unbibium 182		183 Ubc Unbium 183		184 Ubd Unbibium 184		185 Ube Unbeium 185		186 Ubf Unbibium 186		187 Ubg Unbium 187		188 Ubh Unbeium 188		189 Ubi Unbeium 189			

Key

Key

relative atomic mass

Symbol	Name	atomic number	relative atomic mass
A_r	Z		

Surname	Centre Number	Candidate Number
First name(s)		0


GCSE – CONTINGENCY

3410U40-1



Z22-3410U40-1

THURSDAY, 23 JUNE 2022 – MORNING

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5(a)** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	10	
3.	9	
4.	8	
5.	10	
6.	9	
7.	9	
8.	9	
9.	11	
Total	80	

 3410U401
01


JUN223410U40101

10. (a) The list below shows part of the reactivity series.

sodium
aluminium
(carbon)
tin
copper
silver

- (i) Tin is extracted from its ore by heating with carbon. Aluminium is extracted from its ore using a different method. Give the name of the method used to extract aluminium. [1]

.....

- (ii) The equation shows the extraction of tin from tin oxide using carbon.



Tick (✓) the box next to the correct statement. [1]

Carbon is reduced

☐

Tin is oxidised

☐

Tin oxide is reduced

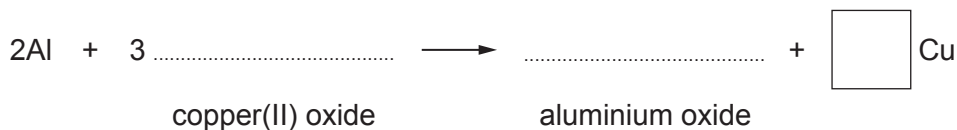
☐

Carbon dioxide is oxidised

☐

- (iii) When aluminium and copper(II) oxide are heated together, aluminium oxide and copper are formed.

Complete and balance the equation for this reaction. [3]



(b) A teacher wanted to find out the position of four metals **A**, **B**, **C** and **D** in the reactivity series.

She heated each metal in turn with oxides of the other three. The results were as follows.

A reduced the oxide of **C**

B reduced the oxide of **A**

B reduced the oxide of **C**

D reduced the oxide of **B**

Place the metals in order of reactivity. [2]

Most reactive

.....

.....

Least reactive

END OF PAPER

7



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

1 H Hydrogen 1															4 He Helium 2																				
7 Li Lithium 3		9 Be Beryllium 4										11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10													
23 Na Sodium 11		24 Mg Magnesium 12										27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18													
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		127 I Iodine 53		131 Xe Xenon 54			
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410U20-1



S23-3410U20-1

MONDAY, 22 MAY 2023 – MORNING

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry
FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 4(b) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	8	
3.	9	
4.	12	
5.	12	
6.	10	
7.	9	
8.	11	
Total	80	

3410U201
01

JUN233410U20101